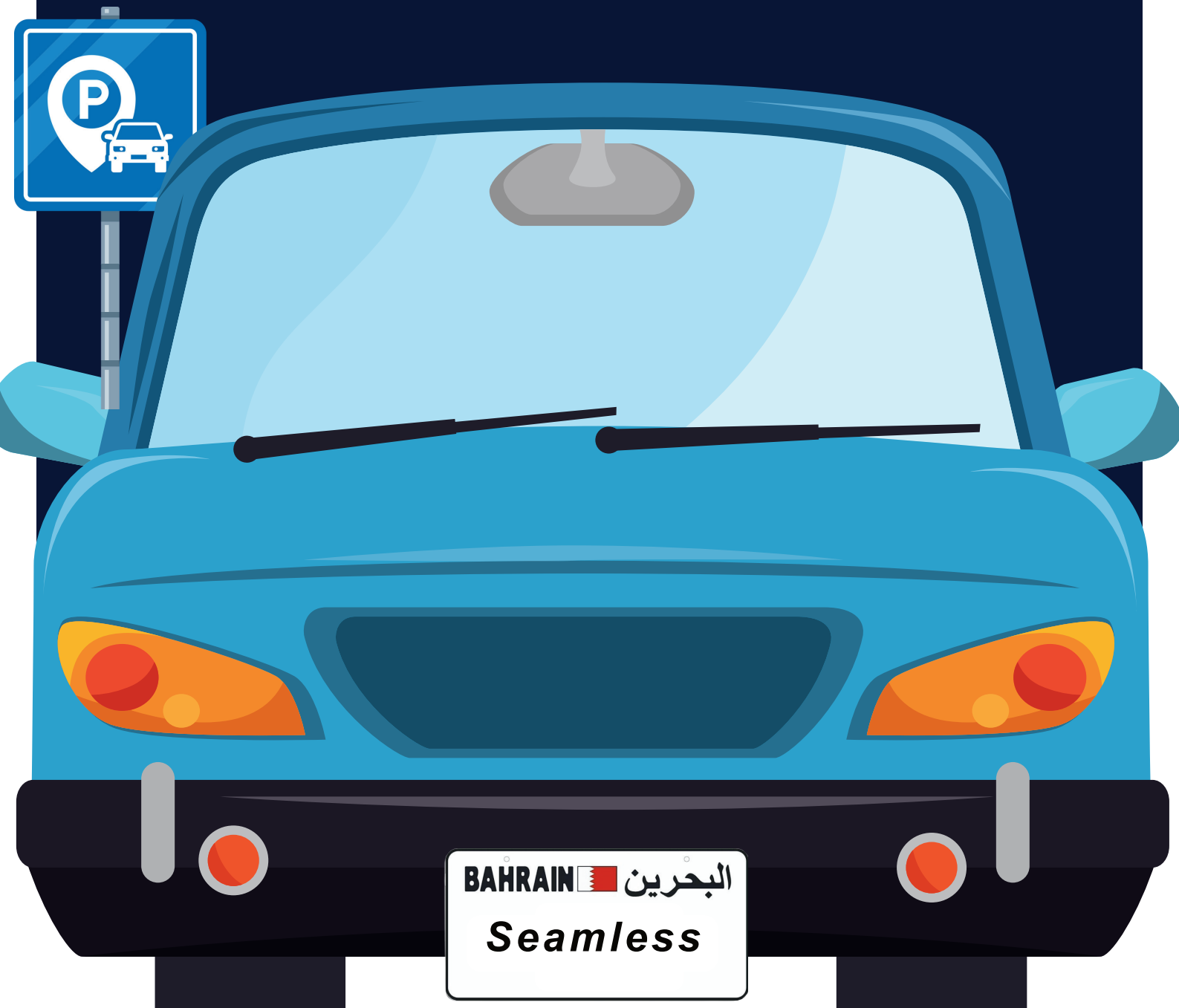
Project by
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PARKATY

Bahrain's innovative

Open Space Car Parking Management Project



TRANSFORMING PARKING WITH INNOVATIVE TECHNOLOGY

PROFILE

In collaboration with the AWS Cloud Innovation Centre at Bahrain Polytechnic,

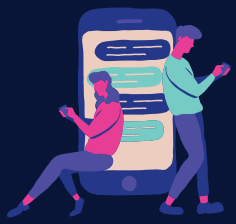
Presenting *Parkaty* - an open-space car parking solution designed to revolutionize the way you park in Bahrain! Say goodbye to traditional parking hassles and embrace a seamless, digitalized experience.

KEY FEATURES



Seamless AI and IoT Integration

Parkaty redefines parking with advanced AI and IoT. Solar-powered cameras, microcontrollers, and sensors enable real-time data processing, automatic number plate recognition, and dynamic parking monitoring for an intelligent and responsive system.



User-Centric Mobile App

Parkaty's app delivers unmatched convenience. Manage tickets, find your car, and handle account details effortlessly. The app ensures seamless digital transactions, eliminating the need for physical ticketing and enhancing your overall parking experience.



Real-Time Analytics Dashboard

Parkaty's dashboard offers precise insights. Administrators access real-time data on parking metrics, user management options, and revenue insights. Powered by advanced AI, it ensures efficient and transparent parking management.

PARKATY'S OPERATIONAL EXCELLENCE: A COMPREHENSIVE OVERVIEW

Discover the inner workings of Parkaty, where seamless operation meets cutting-edge technology. Our comprehensive overview outlines how Parkaty optimizes the parking experience through a harmonious blend of hardware and cloud-based services.

HARDWARE FOUNDATION

Parkaty begins with a Raspberry Pi, armed with a camera module and ultrasonic sensor, forming the backbone of our hardware setup.

IOT CONNECTIVITY

These components connect to an IoT service, initiating the transmission of critical data—timestamp, entry/exit details, device ID, and license plate information—to an S3 bucket, marking the entry event.

AI-DRIVEN INSIGHTS

The entry event triggers image transmission to an EC2 instance hosting a custom-trained AI model. This model efficiently detects license plate types, updates metadata, and initiates follow-up events for database records.

REAL-TIME USER UPDATES

The entry event triggers image transmission to an EC2 instance hosting a custom-trained AI model. This model efficiently detects license plate types, updates metadata, and initiates follow-up events for database records.

EFFICIENT EXIT PROCESS

For exit events, a Lambda function updates database records, deducts the appropriate amount, and signals the WebSocket about the ticket closure. The UI promptly reflects the closed ticket, discreetly stopping the timer.



PARKATY: YOUR PARKING SOLUTION QUESTIONS ANSWERED!



1. HOW DOES PARKATY MAKE PARKING EFFORTLESS?

Parkaty utilizes cutting-edge technology with AI and IoT integration to monitor parking spaces in real-time. This eliminates traditional parking hassles and enhances the overall experience.

2. WHAT SETS PARKATY APART IN THE MARKET?

Parkaty leads with a dynamic blend of cutting-edge tech and user-centric design. Smart sensors, fueled by AI, deliver real-time monitoring for efficient parking. The user-friendly app simplifies space navigation, while the AWS-backed analytics dashboard enhances transparency. With unmatched scalability, speed, and affordability, Parkaty stands out as a leader in urban mobility solutions.

3. WHAT TECHNOLOGIES COMPRISE PARKATY'S SOLUTION?

Parkaty's tech stack, including TypeScript and React Native for the app, TypeScript and Node.js for backend logic, SQL for databases, React for analytics, and Python for machine learning, showcases our commitment to cutting-edge technology, ensuring a user-friendly and advanced urban parking experience.

4. WHAT HAPPENS IF I EXCEED THE PARKING PAYMENT TIME LIMIT?

The parking payment time limit varies based on the specific parking facility and local regulations. Parkaty will notify you through the app and your contact details if you exceed the limit. It's crucial to be aware of the specified time limit to avoid inconvenience and legalities.

5. CAN I EXTEND MY PARKING DURATION REMOTELY OR THROUGH THE APP?

You do not need to extend your parking duration manually; our system automatically tracks the time you spend parking, eliminating the need for users to worry about extending their parking duration through the app.

